

Complete Technology Used :

Technology	Model/Library	Purpose
Python	Programming Language	Entire project
PyTorch	Deep Learning Framework	Model inference
CLIP	Vision Foundation Model	Image embedding generation
FAISS	Vector Database	Similarity search
BLIP	Vision-Language Model	Image caption generation
LLM (Gemma/Llama)	Large Language Model	Complementary recommendation
Prompt Engineering	GenAI Technique	Guide the LLM
Transformers	HuggingFace	Load pretrained models
Pandas	Data Processing	Product metadata
NumPy	Numerical Operations	Embeddings
Matplotlib	Visualization	Display recommendations
PIL	Image Processing	Read product images

models :

1. BLIP :

Model :

`Salesforce/blip-image-captioning-base`

Purpose :

This model is used for generating descriptive captions for product images. These captions help in understanding the visual content of the products.

2. CLIP :

Model :

openai/clip-vit-base-patch32

Purpose :

This model generates normalized image embeddings, which are numerical representations of the images. These embeddings are crucial for finding visually similar products.

3. LLM :

Model

Qwen/Qwen2.5-3B-Instruct

Purpose :

This is a Large Language Model loaded with 4-bit quantization. Its purpose is to generate text-based complementary product recommendations based on a given product.

4. Sentence Transformer :

Model :

all-MiniLM-L6-v2

Purpose :

This model is used to generate embeddings for text queries, specifically for the LLM's recommendations. These text embeddings enable efficient text-based retrieval using a FAISS index.

3. FAISS :

Technology :

Facebook AI Similarity Search

Purpose :

This technology is used to store all embeddings.

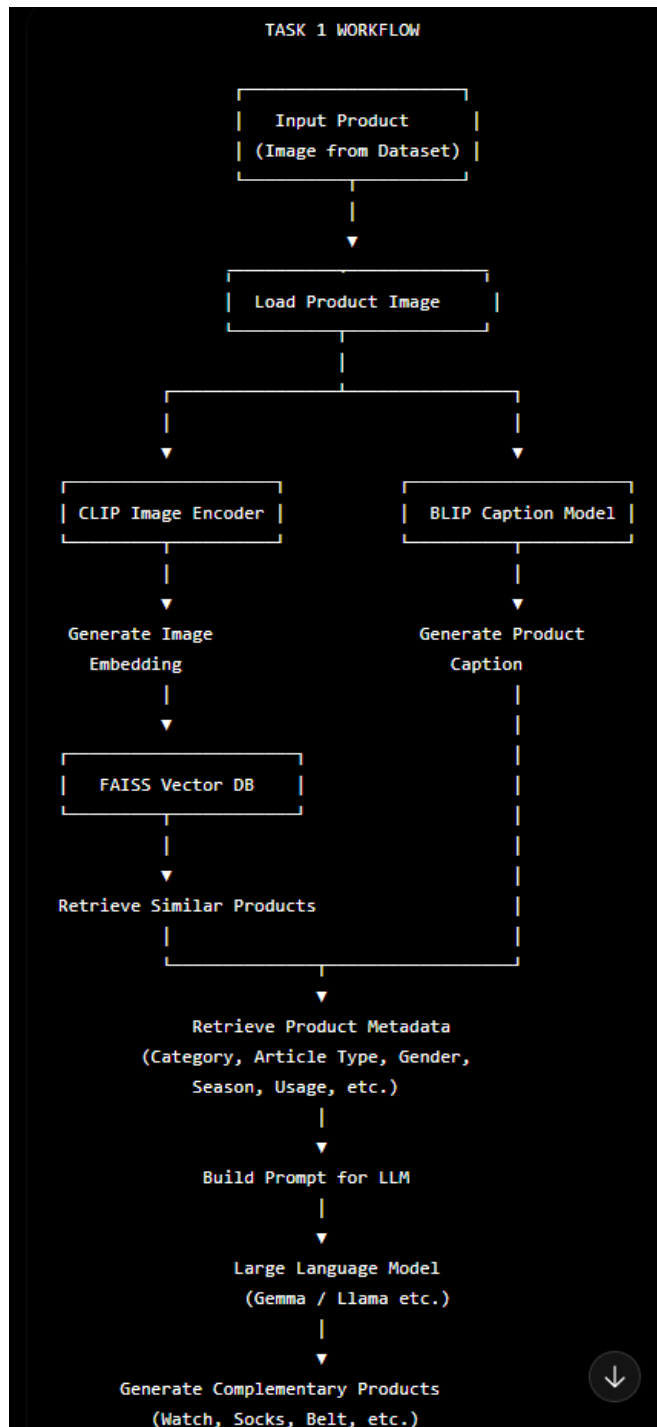
Task 1: Smart Product Recommendation Engine

An intelligent fashion stylist pipeline that recommends complementary outfits based on an initial item.

Technologies Used

- **CLIP**
- **FAISS**
- **Phi-3/Qwen LLM**
- **SentenceTransformer**

Work flow:



Query:

Uploaded Image



Output:



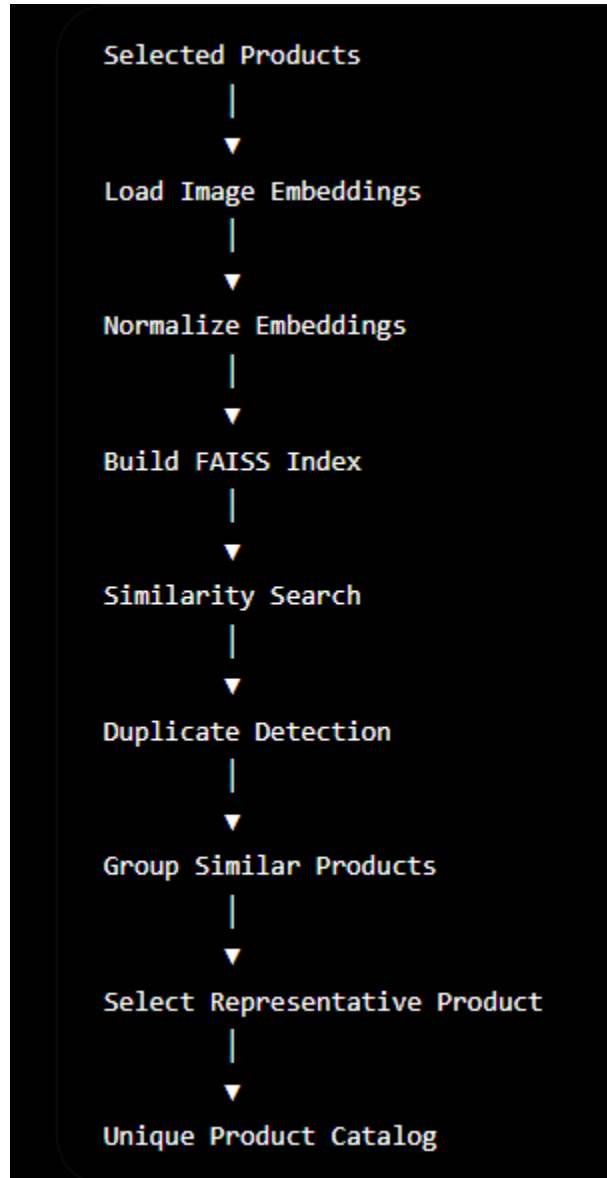
Task 2: Unique Product Catalog Creation

Automates catalog curation by isolating unique product variations and removing duplicate listings.

Technologies Used

- Precomputed Image Embeddings
- FAISS
- Cosine Similarity
- Pandas

Work flow:



Input:

```
Selected images from 5000 to 70000
```

Output:

```
Selected total Products : 65000  
Unique Products       : 7551  
Duplicates Removed: 57449
```

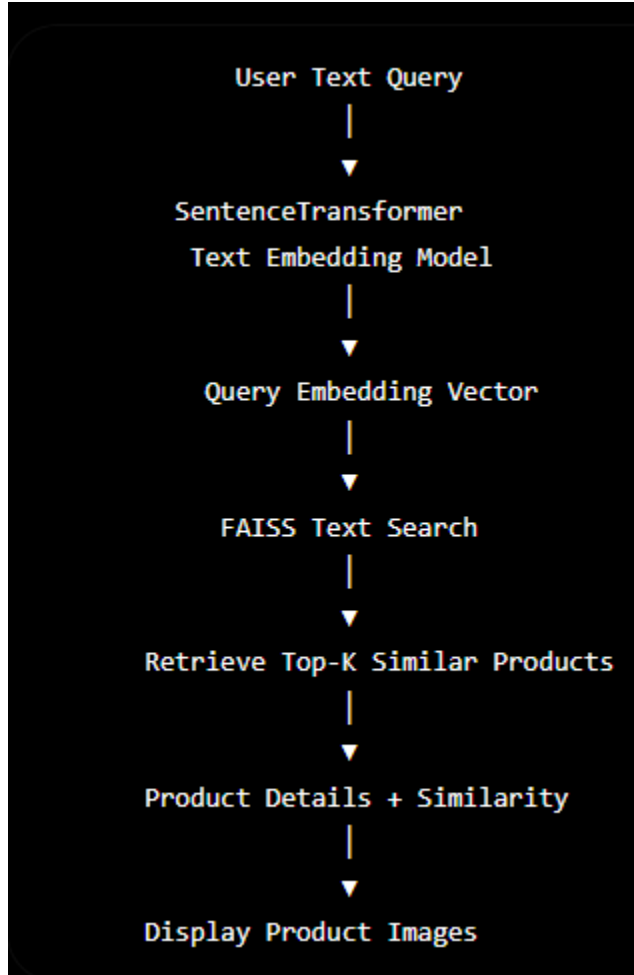
Task 3: Instead of searching using an image, allow users to search using text.

Automates catalog curation by isolating unique product variations and removing duplicate listings.

Technologies Used

- **SentenceTransformer**
- **FAISS**
- **Precomputed Text Embeddings**
- **Matplotlib**

Work flow:



Input:

```
result = search_products("yellow shoes")
```

```
Query : yellow shoes
```

Output:

	productDisplayName	articleType	gender	usage	Similarity	
30210	ID Men Yellow Casual Shoes	Casual Shoes	Men	Casual	0.799825	
13015	Nike Men's 360 Enganche Yellow Shoe	Sports Shoes	Men	Sports	0.771075	
20709	Enroute Girls Yellow Shoes	Casual Shoes	Girls	Casual	0.759719	
24951	ADIDAS Men's Deep Black Yellow Shoe	Sports Shoes	Men	Sports	0.759342	
32835	ID Men Yellow & Black Casual Shoes	Casual Shoes	Men	Casual	0.758143	
	ID Men Yellow Casual	Nike Men's 360 Engan	Enroute Girls Yellow	ADIDAS Men's Deep Bl	ID Men Yellow & Blac	
						